MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) FIFTH SEMESTER

MAJOR ELECTIVE - FOOD MICROBIOLOGY (MG5DSECND301)

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A
Multiple Choice Questions
Answer all questions.
Each question carries 1 mark

1. The lag phase of microbial growth is characterized by: [U][1]
   A) Rapid cell division B) No cell division, adaptation period
   C) Equal cell growth and death D) Decline in cell numbers
2. Which of the following foods is classified as highly perishable? [U][1]
   A) Cereals B) Milk and meat
   C) Dried fruits D) Canned vegetables
3. Which of the following is a common cause of contamination in cereals? [K][2]
   A) Glass fragments B) Mycotoxins from molds
   C) Excess salt D) High protein content
4. Which of the following is an example of chemical spoilage in cereals? [U][2]
   A) Insect infestation B) Rancidity of stored fats/oils
   C) Growth of bacteria D) Presence of stones
5. Which microorganism is most dangerous in canned foods? [U][3]
   A) Clostridium botulinum B) Salmonella typhi
   C) Escherichia coli D) Pseudomonas aeruginosa
6. Spoilage of butter is mainly due to: [K][3]
   A) Rancidity of fats B) Growth of molds

C) Presence of preservatives D) Excess salt content

7. Cross-contamination in meat products occurs when: [U][3]

   A) Raw meat juices contact cooked foods B) Meat is frozen immediately

   C) Meat is vacuum packed D) Meat is pasteurized

8. Fish spoilage is rapid because of: [U][3]

   A) High carbohydrate content B) High water activity and psychrophilic bacteria

   C) Thick protective scales D) Presence of preservatives

9. Which microorganism is commonly used for SCP production? [U][4]

   A) Saccharomyces cerevisiae B) Salmonella typhi

   C) Clostridium botulinum D) Vibrio cholerae

10. Invertase enzyme converts: [K][4]

    A) Glucose into ethanol B) Sucrose into glucose and fructose

    C) Protein into peptides D) Starch into maltose

Part B
Short Answer Questions
Answer any 5 questions
Each question carries 2 marks

11. Classify foods based on perishability with one example for each category? [K][1]
12. Why is sugar less prone to microbial spoilage? [U][2]
13. Mention two molds responsible for spoilage in bakery products. [U][2]
14. Why is poultry highly perishable? [U][3]
15. What is the main cause of milk spoilage? [U][3]
16. Name two microorganisms used for SCP production [U][4]
17. What is invertase? [K][4]

Part C
Short Essay Questions
Answer any 6 questions
Each question carries 5 marks

18. Elaborate on the extrinsic factors affecting the growth of microorganisms in food systems? [U][1]
19. Discuss the common sources of microbial contamination in sugar and sugar products? [U][2]

20. Explain the principles and optimal conditions required for the safe storage of vegetables? [U][2]
21. Explain contamination and spoilage of canned foods? [U][3]
22. Describe the signs and symptoms of spoiled milk. [U][3]
23. Explain the contamination of eggs? [U][3]
24. How is amylase industrially produced using microorganisms? [U][4]
25. Describe the uses of pectolytic enzymes in food industry? [An][4]

Part D
Essay Questions
Answer any 2 questions 10 marks

26. Draw and explain the typical bacterial growth curve in a closed system (batch culture), detailing the four distinct phases? [U][1]
27. Describe the microbial spoilage of fruits. Explain the important spoilage microorganisms and changes during spoilage? [A][2]
28. Explain the contamination and spoilage of meat? [U][3]
29. Discuss the production and Industrial applications of microbial invertase? [U][4]